

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Speech-to-Text Transcription	Converts uploaded audio explanation into text	transcription.py	Done	
2	Semantic Similarity Scoring	Compares transcript meaning to reference concept using Sentence-BERT	semantic_analysis.py	Done	
3	Audio Fluency Analysis	Detects filler words, pause ratio, RMS energy	audio_features.py	Done	
4	Combined Scoring Engine	Merges semantic + fluency into final score and label	scoring.py	Done	
5	AI-Generated Feedback	Gemini-generated natural-language feedback paragraph	feedback_generator.py	Done	
6	Waveform Visualization	Displays audio waveform on dashboard	app.py	Done	
7	PDF Report Generation	Downloadable structured report with scores, transcript, feedback, waveform	report_generator.py	Done	
8	Interactive Dashboard	Streamlit UI with concept selection, upload, results display	app.py	Done	

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Completion Percentage	100%

Additional / Nice-to-Have Features	Custom dark-themed UI design (beyond default Streamlit styling)
Features Tested & Verified	All 8, using real recorded audio samples

Feature Category Breakdown:

S. No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Streamlit dashboard, concept dropdown & audio uploader, waveform visualization
2	Backend / Logic	4	Speech-to-text transcription, semantic similarity scoring, audio fluency analysis (filler words, pauses, RMS), combined scoring engine
3	Database / Storage	1	Reference concepts stored in a structured JSON file (reference_concepts.json)
4	API / Integration	2	Gemini API integration for AI-generated feedback, OpenAI Whisper model integration for transcription
5	Security / Authentication	1	API key managed securely via environment variables / Hugging Face Secrets (not hardcoded in code)